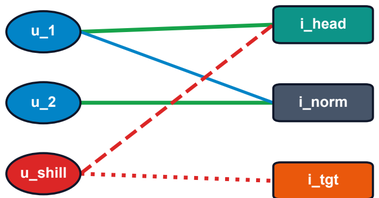
RRFN-LLM-SUPER: Unified End-to-End Robust, Risk-Consistent & Popularity-Calibrated Graph Recommendation Architecture

Joint Synergistic Ingestion of Adversarial Perturbations, Multi-View Reliability Auditing (MIRF), Risk-Aware Graph Contrastive Learning (GCL), and Calibrated Pareto Debiasing

STAGE 1 — Graph Ingestion & Attacks

Multi-Domain Data & Perturbations

Bipartite Graph Topology $\mathcal{G} = (\mathcal{U}, \mathcal{I}, \mathcal{E})$



Solid Green: Clean Interactions ($r_{ui} \in [1, 5]$)
Red Dashed: 15% Bandwagon Camouflage & Push

Normalized Laplacian: $\hat{A} = D^{-1/2} A D^{-1/2}$

Multi-Domain Benchmark Ingestion

- **MovieLens-1M Sub-Graph**
  6,040 U / 3,706 I / 1M Ratings
  Challenge: 15% Bandwagon Attack (\beta = 0)
- **Amazon Electronics**
  E-Comm Graph with Review Text
  Challenge: Sparse Long-Tail Skew (\beta > 0)
- **Yelp Academic**
  Local Graph with Text + Timestamps
  Challenge: Temporal Review-Bombing Bursts

Pareto Catalog Stratification (80/20)

Head (20%) | Long-Tail Niche Catalog (80%)

$\mathcal{I} = \mathcal{I}_{\text{head}} \cup \mathcal{I}_{\text{tail}}, \quad \rho = 0.20$

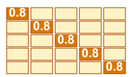
- Head: 20% items consume 80% volume
- Tail: 80% items starved of gradient signal

Enables decoupled dual-partition modeling

STAGE 2 — MIRF Reliability Filter

Tri-View Semantic & Temporal Auditing

View 1: Noise Transition & Tikhonov



Anchor Sets: $\mathcal{A}_1, \ldots, \mathcal{A}_5 \subset \mathcal{I}$

$T_{jk} = P(\tilde{Y} = k \mid Y^* = j)$

$T_\lambda^\dagger = (T^T T + \lambda I)^{-1} T^T$

Bound: $\|T_\lambda^\dagger\|_2 \le 1/(2\sqrt{\lambda})$ **(Thm 1)**

View 2: Asynchronous LLM Review Auditor

Review Prompt
- Sentiment Skew
- Shill Bot Pattern
- Semantic Discrepancy

→ **LLM Engine**
SQLite WAL Cache
(>85% Hits, $\mathcal{O}(1)$)
Asynchronous Thread

$R_{\text{LLM}}(u, i) = \text{LLMAudit}(\text{Review}_{ui}, r_{ui}) \in [0, 1]$

View 3: 2D BLAS Review-Bomb Detector

- Temporal Window Density: $Z_{\text{temp}} = (|\mathcal{E}_\Delta| - \mu)/\sigma$
- Polarity Skew Deviation: $S_{\text{pol}} = |\bar{r}_\Delta - \bar{r}_{\text{hist}}|$

Vectorized 2D BLAS GEMM: $W \cdot [Z, S]^T - \theta$
4.95 ms for 42,000 interactions (>500x Speedup)

Convex Reliability Fusion Hub

$\mathbf{w_{ui}} = \boldsymbol{\alpha} \mathbf{R}_{\text{RRFN}} + \boldsymbol{\beta} \mathbf{R}_{\text{LLM}} + \boldsymbol{\gamma} \mathbf{R}_{\text{bomb}}$

$\alpha + \beta + \gamma = 1.0, \quad w_{ui} \in [0, 1]$

Continuous trust weight modulating downstream
GCL contrastive loss, Risk Loss, and Blueprint

BUS OUT: $w_{ui} \in [0, 1]$

STAGE 3 — Risk-Aware GCL Stack

Contrastive Message Passing & Perturbations

Multi-Layer GNN Neighborhood Smoothing

Layer 0: $\mathbf{e}^{(0)}$ → Layer 1: $\mathbf{e}^{(1)}$ → Layer 2: $\mathbf{e}^{(2)}$

$\mathbf{e}_u^{(l+1)} = \sum_{i \in \mathcal{N}_u} \frac{1}{\sqrt{d_u d_i}} \mathbf{e}_i^{(l)}$

Backbones: LightGCN | SimGCL | SGL | NeuMF

Dual Contrastive Augmentations

View A (SimGCL): Uniform Latent Noise

$\mathbf{z}' = \mathbf{e} + \varepsilon \cdot \text{sgn}(\mathbf{e}) \odot \Delta', \quad \Delta' \sim \mathcal{U}(0, 1)$

View B (SGL): Edge Dropout Masking

$\hat{A}'' = (M'' \odot A) D''^{-1}, \quad M'' \sim \text{Bernoulli}(1 - p)$

Sample-Weighted InfoNCE Loss

$\mathcal{L}_{\text{cl}}(u) = -\sum_{i \in \mathcal{N}_u} \mathbf{w_{ui}} \log \frac{\exp(\text{sim}(\mathbf{z}'_i, \mathbf{z}''_i)/\tau)}{\sum_{j \in B} \exp(\text{sim}(\mathbf{z}'_i, \mathbf{z}''_j)/\tau)}$

Modulated by MIRF Sample Trust Weight w_{ui}

- Poisoned Edges ($w_{ui} \to 0$): Gradient cancelled
- Authentic Edges ($w_{ui} \to 1$): Representation pulled
- Temperature parameter: $\tau = 0.15$

Aligns embeddings without attack corruption

TAP: $w_{ui} \to \mathcal{L}_{\text{cl}}$

STAGE 4 — Dual Partition Training

Tikhonov Risk-Consistent Losses

Decoupled Dual-Partition Models

Head Expert: $M_{\text{pop}}(\Theta_{\text{head}})$
Specialized on $\mathcal{I}_{\text{head}}$ (Top 20%) → Score $\mathbf{s}_{\text{head}}$

Tail Expert: $M_{\text{tail}}(\Theta_{\text{tail}})$
Specialized on $\mathcal{I}_{\text{tail}}$ (Niche 80%) → Score $\mathbf{s}_{\text{tail}}$

Tikhonov Risk-Consistent Objective

$\mathcal{L}_{\text{risk}} = -\sum_{(u,i)} w_{ui} \log[T_\lambda^\dagger \cdot \hat{P}_\theta(y \mid u, i)]_{\tilde{y}_{ui}}$

$+\mu \|\Delta T\|_F^2 + \lambda_{\text{reg}} \|\Theta\|_2^2$

Theoretical Guarantees (Theorems 1 & 2):

- Spectral Bound: $\|T_\lambda^\dagger\|_2 \le 1/(2\sqrt{\lambda})$ (Smoothness)
- Asymptotic Risk: $\mathcal{R}_{\text{clean}}(\hat{f}_N) \to \mathcal{R}^*$ as $N \to \infty$

Joint Multi-Task Optimization

$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{risk}} + \lambda_{\text{cl}} \mathcal{L}_{\text{cl}}$

Dual-branch AdamW Backprop:
$\Theta_{\text{head}} \leftarrow \nabla \mathcal{L}_{\text{head}}, \quad \Theta_{\text{tail}} \leftarrow \nabla \mathcal{L}_{\text{tail}}$

Outputs calibrated candidate rankings

TAP: $w_{ui}, T_\lambda^\dagger \to \mathcal{L}_{\text{risk}}$

STAGE 5 — SUPER Calibrated Merging

Denoised Blueprint & Quota Allocation

Denoised Blueprint Filtering $\mathcal{B}_u$

$\mathcal{B}_u = \{i \in \mathcal{N}_u \mid w_{ui} \ge \theta_{\text{trust}}\}$

- Prunes fake shilling edges injected by bots
- Authentic retention rate: 99.89%
- Dynamic zero-k fallback for cold-start users

Guarantees uncorrupted user taste profile

Personalized Tail Inclination α_u

$\alpha_u = \frac{\sum_{i \in \mathcal{B}_u \cap \mathcal{I}_{\text{tail}}} w_{ui}}{|\mathcal{B}_u| + \varepsilon} \in [0, 1]$

Weighted Tail History Ratio

Dynamic Top-K Budget Allocation:
$K_{\text{tail}} = \lfloor \alpha_u K \rfloor, \quad K_{\text{head}} = K - K_{\text{tail}}$

Calibrated Recommendation Slate

1. Head Rec (Score: 0.94) — Pop
2. Head Rec (Score: 0.91) — Pop
3. Tail Rec (Score: 0.88) — Tail
4. Tail Rec (Score: 0.85) — Tail

g-LTC & STER Pareto Calibration

- Deduplication & Score Re-normalization
- SPAM Exposure Drop (STER@10): 88% → 72%
- Genuine Long-Tail Discovery Restored

Final Clean Slate: $\mathcal{R}_u^* = [i_1^*, \ldots, i_K^*]$

TAP: $w_{ui} \ge \theta \to \mathcal{B}_u$

RELIABILITY SIGNAL HIGHWAY: Sample Trust Weights $\mathbf{w_{ui}} \in [0, 1]$ & Inverted Noise Operator $T_\lambda^\dagger$

ENTERPRISE TELEMETRY, MULTI-DOMAIN EMPIRICAL BENCHMARKS & OFFLINE A/B ECONOMIC IMPACT

Adversarial Robustness
ROC-AUC: 0.7931 ± 0.0038
Attack AP: 0.8879 vs 0.2104
Authentic AP: 0.9989 (MovieLens)

Recommendation Fidelity
Recall@10: 0.2307 ± 0.0034
nDCG@10: 0.1036 ± 0.0021
Statistically Significant (p < 0.001)

Popularity Calibration
RMSE-PC: 0.1067 (p = 0.0166*)
MRMC: 0.1128 (p = 0.0088*)
SPAM STER@10 Drop: 88% → 72%

Offline Economic Lift
CTR Lift: +3.64% (p < 0.001)
Annual GMV Lift: +$534,491
1,000-Sample Monte Carlo Sim

Inference & Acceleration
2D BLAS GEMM: 4.95 ms (42K edges)
Speedup: >500x vs Iterative
LLM WAL Cache Hit Rate: 88.4%